Mechanical Properties (Part 1)



MO-201 Materials Science
Odd Semester 2026-27
Dr. S. Sinha (Met. Engg.)

References:
Materials Science & Engg. by V. Raghavan
Fundamentals of Materials Science by W.D. Callister

Why study mechanical properties?

- Materials are subjected to forces or loads when in service
- Deformation
- Fracture or failure
- Mechanical testing for material design

Plastic Deformation and Fracture

- Deformation
  - What is Elastic deformation?
  - What is Plastic deformation?
  - What is Fracture and/or Failure?

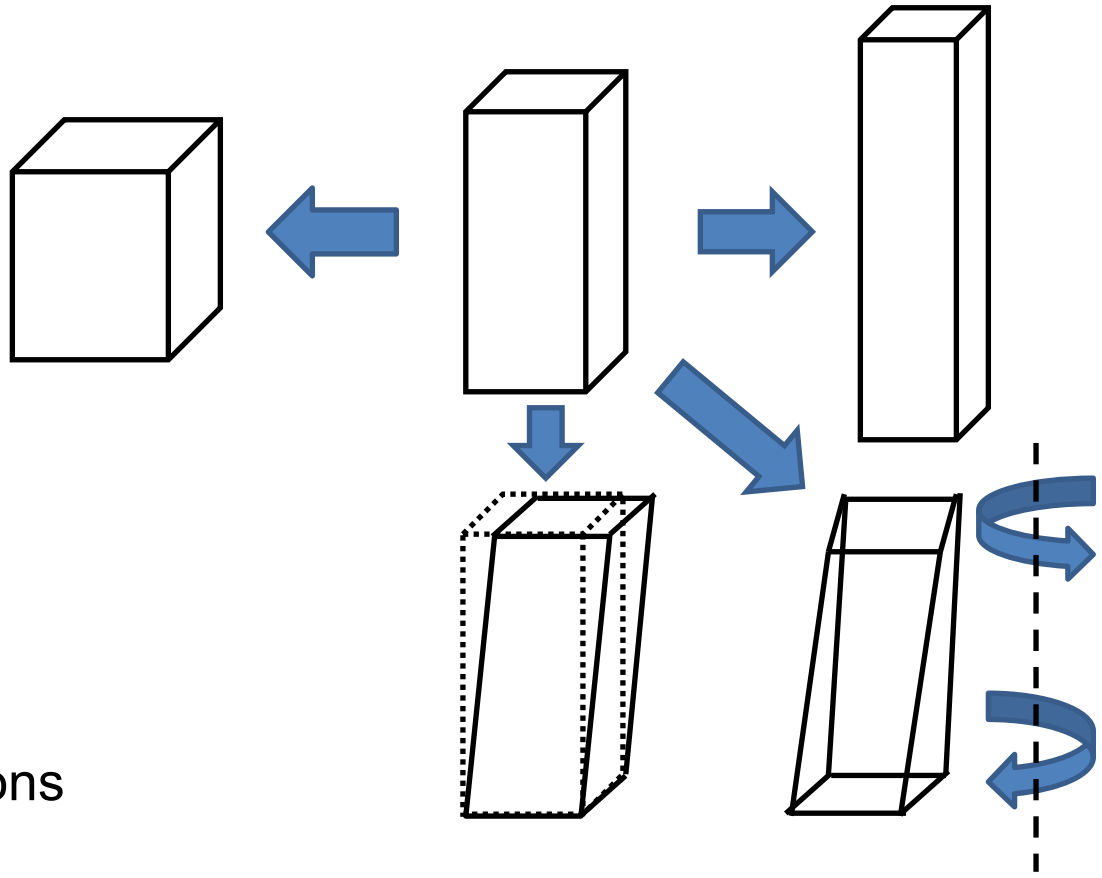
- Suppose a car has an accident
  - Luckily, nothing at all happened
  - Some scratches on the side
  - A dent on the side
  - Unfortunately, the car broke into pieces like they show in the movies

Plastic Deformation and Fracture

- Plastic deformation refers to permanent, non-recoverable change in size or shape
- After a point, the material breaks – Fracture
- How much Elastic/Plastic deformation can the material accommodate before fracture or failure?
- Invoke the concept of stress/strain, which we will constantly talk about in testing

Types of loading

- Tension
- Compression
- Shear (and Torsion)
- Concept of strain
  - Change in dimensions
  - Angular distortion

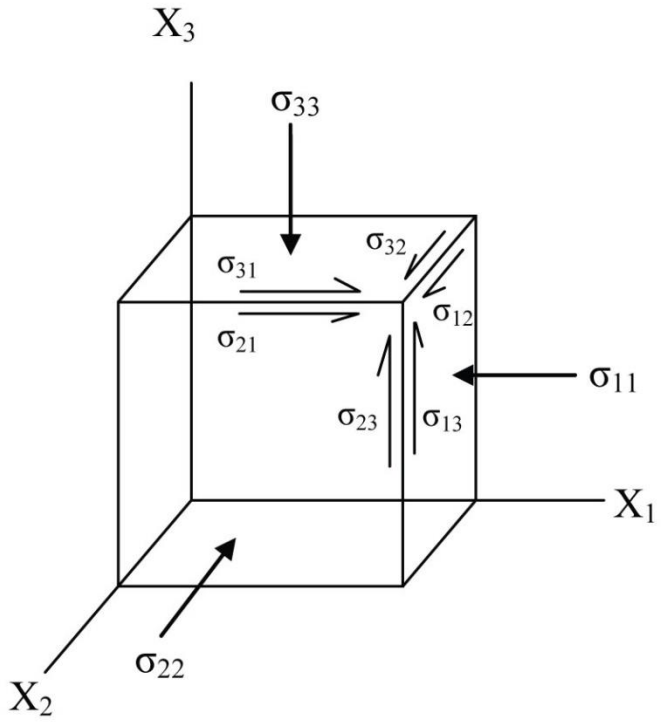


Concepts of stress/strain

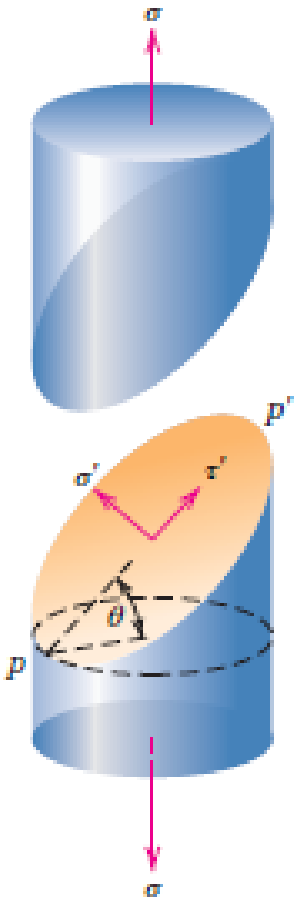
- Simple definitions
  - Stress is the force per unit area
    - Stress = Load (P)/Area of cross-section(A_0)
  - Strain is the change in dimension per unit length
    - Strain = Displacement (ΔL)/Gauge length (L_0))

- But remember, Stress and Strain are actually 2nd Rank Tensors and Elastic stiffness is a 4th Rank Tensor!

Stress tensor



$$\left[\ \sigma_{ij}\ \right] = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix}$$



Normal and Shear components of stress

Examples of mechanical tests

- Hardness test
- Tension test
- Compression test
- Shear and Torsion test
- Three point bend test
- Fatigue test
- Impact test

Key mechanical design properties

- Stiffness
- Strength
- Hardness
- Ductility
- Toughness

What do you mean by Brittle and Ductile?

Relation between Mechanical testing and Deformation mechanisms

Also, there is Creep test - time dependent deformation at elevated temperature under constant stress/load

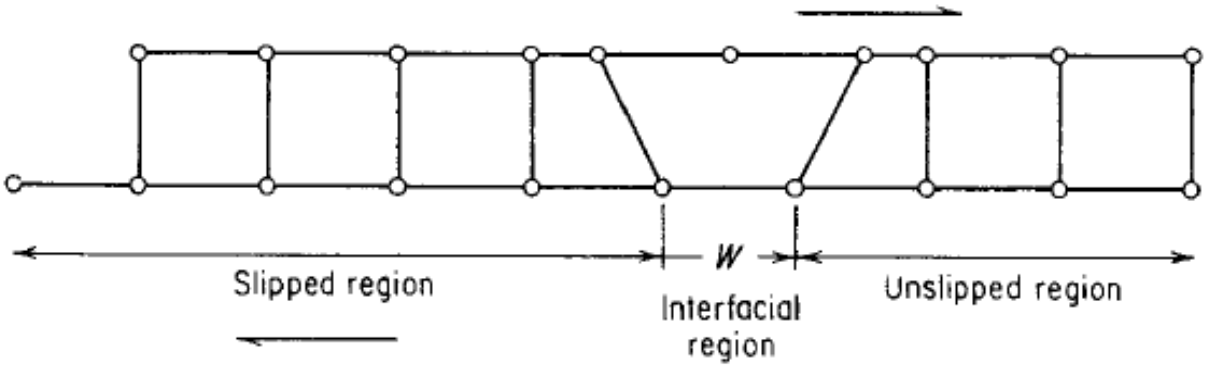
Plastic deformation mechanisms - Slip, Twinning, Climb, Grain boundary sliding, etc.

Schmid factor/CRSS

Glissile/Sessile dislocations

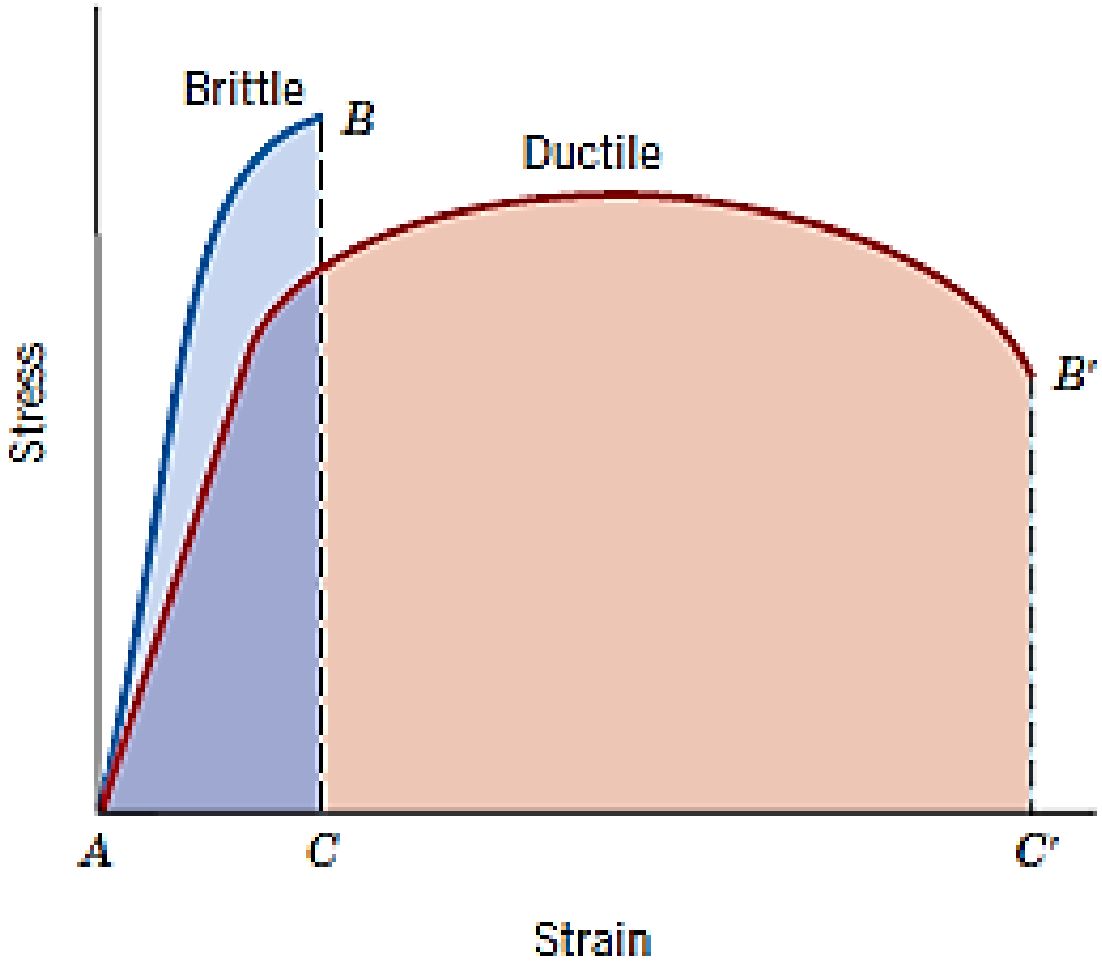
Theoretical shear strength of perfect crystal

Peierls stress/Peierls-Nabarro force to move a dislocation



$$\tau_p \approx \frac{2G}{1-\nu} e^{-2\pi w/b} \approx \frac{2G}{1-\nu} e^{-[2\pi a/(1-\nu)b]}$$

Tensile Test



Tensile stress-strain curve